



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

**CANDIDATE
NAME**

CLASS

CHEMISTRY

9729/03

JC2 Preliminary Examination
Paper 3 Free Response

19 September 2018
2 hours

Candidates answer on separate paper.

Additional materials : Writing Papers

Data Booklet

2 Cover Pages (one for Section **A** and one for Section **B**)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

This document consists of **12** printed pages and **0** blank page.

Section A

Answer **all** the questions in this section.

- 1 Hydrogen sulfide, H_2S , is a colourless gas with the characteristic foul odour of rotten eggs. It is very poisonous, corrosive and flammable. It occurs naturally in volcanic gases and natural gas and is often produced from the microbial breakdown of organic matter.

- (a) H_2S burns readily in oxygen to form water and an acidic gas which briskly decolourises acidified potassium manganate(VII).

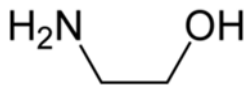
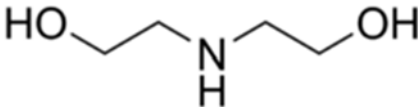
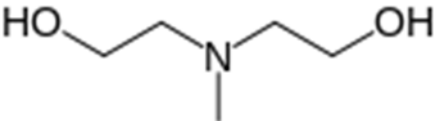
During the combustion process, sulfur atomises and a brightly coloured flame similar to that of its element is observed.

Construct a balanced chemical equation to represent the combustion reaction and predict the colour of the flame.

[2]

- (b) The toxicity of H_2S is comparable with that of what carbon monoxide does to the human body, preventing cellular respiration. It is therefore critical to remove H_2S from raw natural gas. This process, commercially known as “sweetening”, utilises amine treating technologies.

Common “sweetening” agents are as shown below.

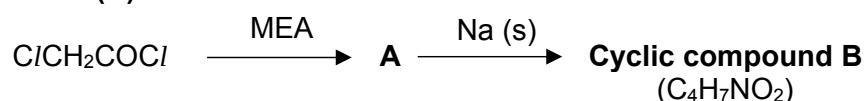
Name	Chemical Structure	pK_b
Monoethanolamine (MEA)		4.55
Diethanolamine (DEA)		5.12
Methyldiethanolamine (MDEA)		5.48

- (i) Describe how H_2S impedes cellular respiratory in the human body. [2]
- (ii) Amine compounds are *weak Bronsted-Lowry bases*. What do you understand by the term “*weak Bronsted-Lowry bases*”? [1]
- (iii) Explain the trend of the pK_b values of MEA, DEA and MDEA. [2]

- (iv) MEA consists of both amine and alcohol functional groups. It can be exploited in successive reactions involving halogeno organic compounds. A series of experiments are carried out to find out the different reactivity of dihalogeno compounds.

Complete the following of the organic syntheses, drawing the structures of the intermediates, products and stating the reagents and conditions where appropriate.

Reaction (A):

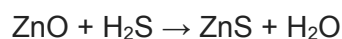


Reaction (B):



[4]

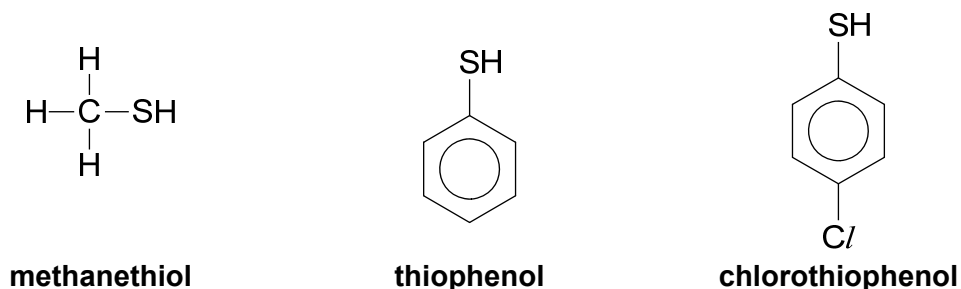
- (c) Zinc oxide is also used to scavenge hydrogen sulfide in natural gas, where the by-product, zinc sulfide, may be easily filtered off.



The solubility of ZnS in pure water is $1.23 \times 10^{-10} \text{ g dm}^{-3}$ at 25°C .

- Write an expression for the solubility product, K_{sp} of ZnS and calculate its value. State its units clearly. [2]
- How would the solubility of ZnS in aqueous $\text{Zn}(\text{NO}_3)_2$ compared to that in pure water? Use *Le Chatelier's Principle* to justify your answer. [2]
- Equal volumes of $0.005 \text{ mol dm}^{-3} \text{ Zn}^{2+}$ solution and $0.002 \text{ mol dm}^{-3} \text{ Na}_2\text{S}$ solution were mixed. Predict if any precipitate could be filtered off this mixture. [1]

- (d) Thiols are a class of organosulfur compounds known to have strong unpleasant odours. For instance, methanethiol, CH_3SH , is one of the main odourants of human faeces, body odour and bad breath. Other examples of common thiol structures are shown below.



As thiols are the sulfur analogue of alcohols, their relative acidities are often compared. One notable difference is the ability of thiols to react with aqueous alkalis.



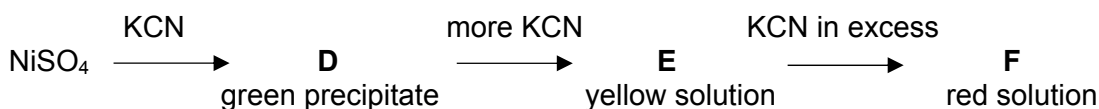
The table below shows the $\text{p}K_{\text{a}}$ values of alcohols and thiols.

Name of compound	$\text{p}K_{\text{a}}$
Butanol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$	15
Phenol, $\text{C}_6\text{H}_5\text{OH}$	10
Buthanetiol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{SH}$	10
Thiophenol, $\text{C}_6\text{H}_5\text{SH}$	6
Chlorothiophenol, $\text{C}_6\text{H}_5\text{Cl/S}$	<i>y</i>

- (i) With reference to the information above, suggest an explanation for the different $\text{p}K_{\text{a}}$ value of butanol as compared to buthanetiol. [1]
- (ii) Predict the value *y* and explain your answer. [2]
- (iii) Calculate the pH of the solution when 15 cm^3 of 0.10 mol dm^{-3} NaOH is added to 25 cm^3 of 0.10 mol dm^{-3} $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{SH}$. [2]

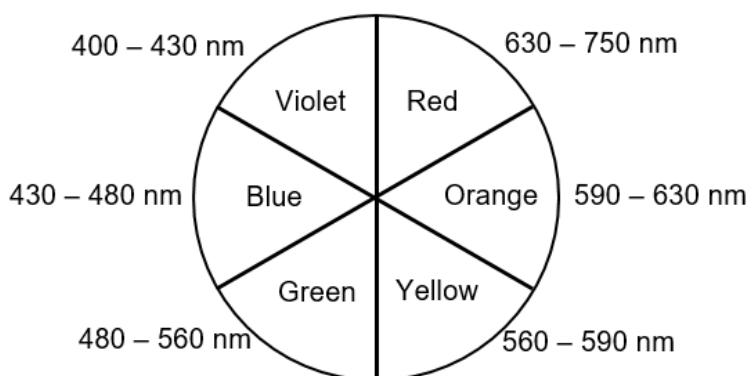
[Total: 21]

- 2 (a) (i) Draw a fully labelled diagram to show how the standard electrode of $\text{Ni}^{2+}(\text{aq})/\text{Ni}(\text{s})$ is being measured in a laboratory. Clearly label the direction of electrons flow and polarity of the electrodes in your diagram. [3]
- (ii) Adding a measured quantity of KCN to a solution of NiCl_2 produces the complex $[\text{Ni}(\text{CN})_2\text{Cl}_2]^x$. Suggest how the value of the electrode potential of the $\text{Ni}^{2+}(\text{aq})/\text{Ni}(\text{s})$ half-cell will vary upon the addition of KCN and deduce the overall charge, x , on this complex. [3]
- (b) Determine the $E^\ominus_{\text{cell}}$ and the $\Delta G^\ominus$ when the $\text{Ni}^{2+}(\text{aq})/\text{Ni}(\text{s})$ half-cell is connected with the $\text{Fe}^{3+}(\text{aq})/\text{Fe}^{2+}(\text{aq})$ half-cell. [2]
- (c) An aqueous solution of KCN is gradually added to a solution of NiSO_4 until the KCN is in excess. The following series of reactions takes place.



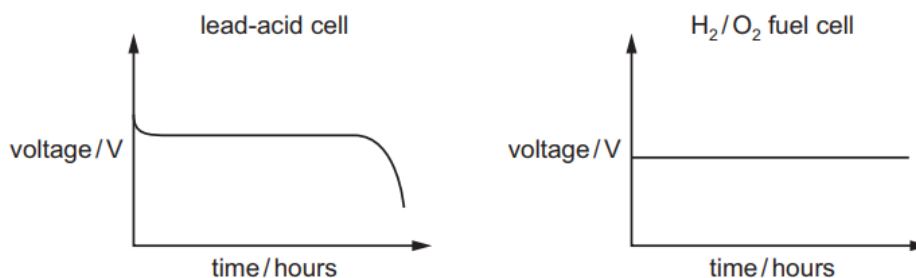
- The oxidation state of nickel does **not** change during the reactions.
- None of **D**, **E** or **F** contains sulfur.
- **D** contains no potassium.
- The K: Ni ratio in **E** is 2:1.
- The K: Ni ratio in **F** is 3:1

- (i) Use the information to suggest the formulae of **D** and **F**. [2]
- (ii) Using the colour wheel provided, suggest if **E** or **F** has a larger energy gap when the d-orbital split into two different energy levels.



[1]

- (d) The diagrams show how the voltage across two different cells changes with time when each cell is used to provide an electric current.



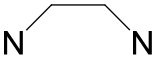
Suggest a reason why.

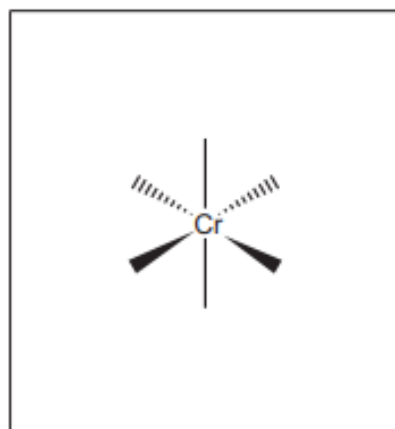
- (i) The voltage of the lead-acid cell changes after several hours. [1]
 - (ii) The voltage of the fuel cell remains constant. [1]
- (e) Describe, using the orientation of the d-orbitals, the splitting of degenerate d-orbitals into two energy levels in octahedral complexes such as [Fe(CN)₆]³⁻ and [Fe(CN)₆]⁴⁻. [4]
- (f) 1,2-diaminoethane, *en*, is a bidentate ligand.

- (i) What is meant by the terms *bidentate* and *ligand*? [2]
- (ii) There are isomeric complex ions with the formula [Cr(*en*)₂Cl₂]⁺.

When all the four nitrogen atoms of the *en* ligands are on the same plane, it is consider a trans configuration. When two of the nitrogen on the *en* ligands are on different plane, it is consider a cis configuration.

Using the three-dimensional diagram below, draw and label the cis-trans isomers on the writing paper you are provided.

(You may use  to represent *en*.)



[2]

[Total: 21]

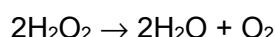
3 (a) Solutions of iron(II) compounds are commonly prepared in the laboratory.

- (i) By selecting two appropriate half equations from the *Data Booklet*, explain why iron(II) solutions are normally stored in the presence of acids instead of bases.

[3]

- (ii) With the aid of suitable equations and standard electrode potential values from the *Data Booklet*, explain why an acidic solution of iron(II) can catalyse the decomposition of hydrogen peroxide.

Decomposition of hydrogen peroxide:



State and explain the type of catalysis involved.

[3]

- (iii) Using an appropriate sketch of the Boltzmann distribution; explain why the addition of acidic solution of iron(II) can catalyse the decomposition of hydrogen peroxide.

[3]

(b) Strontium hydroxide is a strong base and has several industrial applications such as plastic stabiliser and paint drier.

The standard enthalpy change of neutralisation was determined experimentally by mixing known volumes of aqueous hydrochloric acid and aqueous strontium hydroxide in a calorimeter. The following results were obtained:

- Initial temperature = 25.0 °C
- Final temperature = 35.4 °C
- Volume of 2 mol dm⁻³ hydrochloric acid = 50.0 cm³
- Volume of 0.77 mol dm⁻³ of strontium hydroxide = 50.0 cm³
- Heat capacity of calorimeter = 9.43 J K⁻¹

- (i) Define *standard enthalpy change of neutralisation*.

[1]

- (ii) Use the data above to calculate the standard enthalpy change of neutralisation in kJ mol⁻¹.

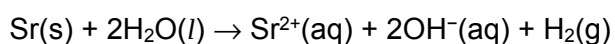
[3]

- (iii) The enthalpy change of neutralisation between aqueous strontium hydroxide and aqueous ethanoic acid was found to be different from the value calculated in **b(ii)**.

State and explain how the magnitude of this value differ from **b(ii)**.

[2]

- (c) Aqueous strontium hydroxide is formed when solid strontium metal reacts with water



By using the following enthalpy changes and your answer in **b(ii)**, draw an energy cycle to calculate the enthalpy change for the above reaction involving strontium and water.

	kJ mol⁻¹
Enthalpy change for $\text{Sr(s)} \rightarrow \text{Sr}^{2+}(\text{g}) + 2\text{e}^{-}$	+1772
Enthalpy change of hydration of strontium ions	-1337
Enthalpy change for $2\text{H}^{+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{H}_2(\text{g})$	-850

[3]

Total [18]

Section B

Answer **one** question in this section.

4 (a) Four samples of Period 3 chlorides, **G**, **H**, **J** and **K** are examined.

- These chlorides reacted completely in water, leaving no precipitates. These solutions were tested with universal indicator paper. **G**, **J** and **K** were found to be acidic, while **H** was found to be neutral.
- The melting points of **H** and **J** are approximately 10 times higher than the melting points of **G** and **K**.
- Solid anhydrous **G** is required in the reaction of phenylmethanol with chlorine to form C_7H_7OCl .
- Solid anhydrous **K** reacts with phenylmethanol to form C_7H_7Cl at room temperature.

- Identify the chlorides **G**, **H**, **J** and **K**. [2]
- Explain why **H** forms a neutral solution. [2]
- Draw the structure of the resulting compound, C_7H_7OCl . [1]
- Draw the structure of the resulting compound, C_7H_7Cl . [1]
- Suggest why chloride **K** has to be anhydrous for the reaction between chloride **K** and phenylmethanol to occur. Write a chemical equation to support your answer. [2]
- Using structure and bonding, explain why chloride **J** has a melting point approximately 10 times higher than chloride **G** and **K**. [1]

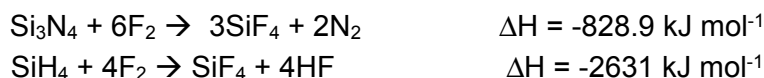
(b) The reaction of silicon tetrachloride with moist ethoxyethane, produces either oxochlorides, Si_2OCl_6 or $Si_3O_2Cl_8$.

When 0.10g of one of the oxochloride completely reacted with water, all of its chlorine was converted into chloride ions, and produced 0.303 g of silver chloride precipitate when an excess of aqueous silver nitrate was added.

Deduce the identity of the oxochloride. [2]

- (c) Fluorine bomb calorimetry has been used to investigate inorganic substances such as silicon compounds that cannot be completely burned in conventional calorimetric reagents such as oxygen under high pressure.

The enthalpy change of reaction of the following silicon containing compounds were investigated during a fluorine bomb calorimeter.



The following enthalpy change of formation is also given below.

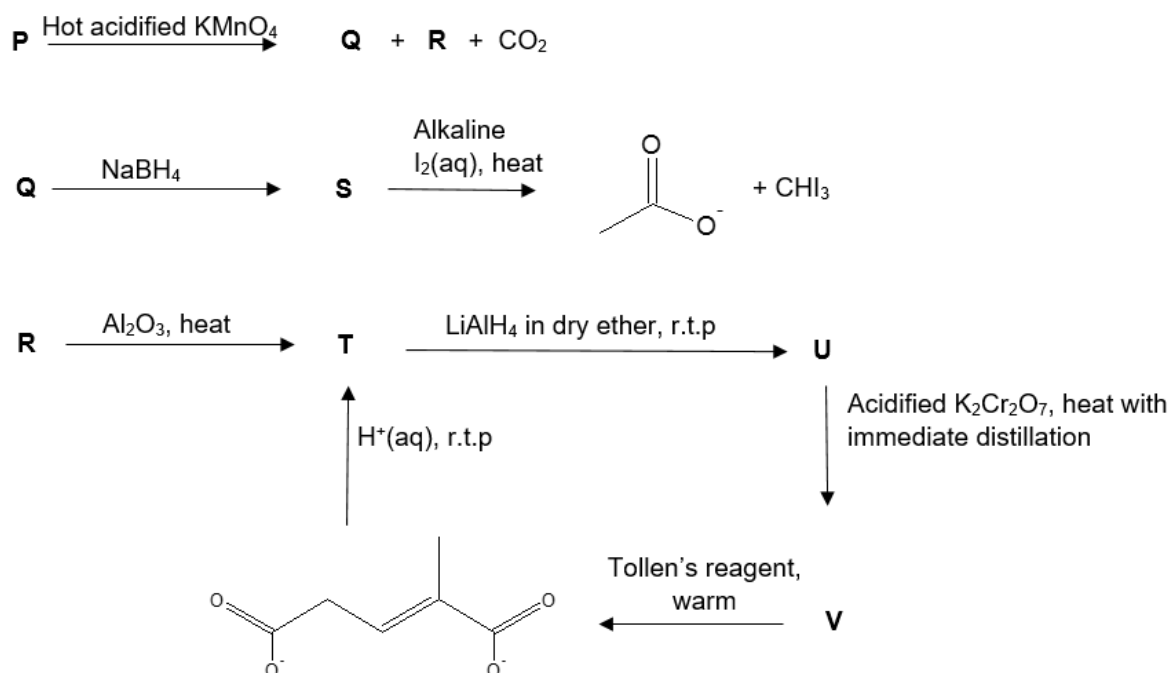
Compound	$\Delta H_f / \text{kJ mol}^{-1}$
HF(g)	-272.6
Si ₃ N ₄ (s)	-3686.1

Calculate the standard enthalpy change of formation of SiF₄.

Hence, calculate the standard enthalpy change of formation of SiH₄.

[2]

- (d) Linalool, **P**, is a compound which gives the sweet scent of lavender. Its structure can be deduced from the following series of reactions.

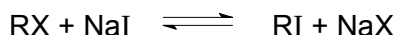


- (i) Draw the structures of **S** and **Q**. [2]
- (ii) Draw the structures of **R**, **T**, **U** and **V**. [4]
- (iii) Hence, deduce the structure of linalool, **P**. [1]

[Total: 20]

5 This question is about the reactions of halogen containing organic compounds.

- (a)** The Finkelstein reaction, named after the German chemist, Hans Finkelstein, involves the replacement of the halogen atom in a halogenoalkane by an iodide ion from a solution of sodium iodide dissolved in propanone. This reaction goes almost to completion and a precipitate of sodium halide is formed as shown below.



A series of experiment can be carried out to determine the order of reaction with respect to a halogenoalkane, $\text{C}_4\text{H}_9\text{Br}$, and iodide ion. Different concentrations of $\text{C}_4\text{H}_9\text{Br}$ and sodium iodide were used and the following initial rates were obtained.

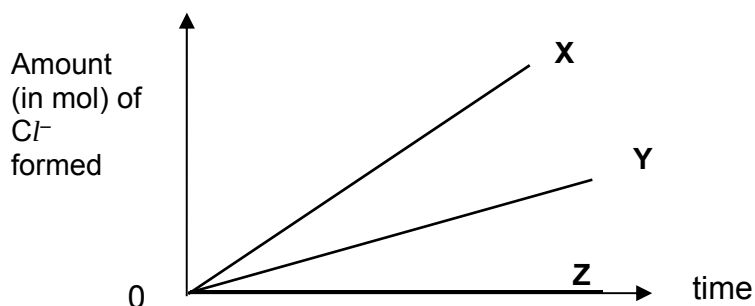
Experiment Number	$[\text{C}_4\text{H}_9\text{Br}] / \text{mol dm}^{-3}$	$[\text{NaI}] / \text{mol dm}^{-3}$	Rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	0.04	0.60	5.68×10^{-5}
2	0.02	0.30	1.42×10^{-5}
3	0.60	0.60	8.52×10^{-4}

- (i)** Use these data to deduce the order of reaction with respect to each reactant, showing how you arrive at your answers. Hence, write a rate equation for the reaction. [3]
- (ii)** Based on your answer in **(a)(i)**, deduce a suitable structure for the halogenoalkane, $\text{C}_4\text{H}_9\text{Br}$ which contains a chiral carbon. [2]
- (iii)** Using your answer in **(a)(i)** and **(a)(ii)**, describe a mechanism that is consistent with your rate equation. Show relevant lone pairs and dipoles, and use curly arrows to indicate the movement of electron pairs. [2]
- (iv)** Sodium chloride is precipitated during the Finkelstein reaction due to its limited solubility in propanone.

Explain why the Finkelstein reaction goes almost to completion despite the C-I bond being weaker than the C-Cl bond. [1]

- (b) Halogen-containing organic compounds can be hydrolysed under suitable conditions to produce the corresponding halide ions.

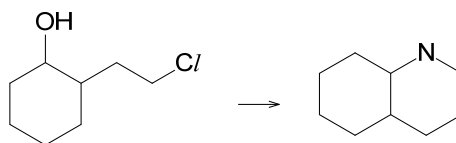
A student investigated the amount of Cl^- produced by hydrolysing ethanoyl chloride, chloroethane and chlorobenzene and the results are shown below.



Deduce **X**, **Y** and **Z**.

[3]

- (c) In no more than 4 steps, outline how the following transformation can be achieved. State the reagents and conditions for each step, as well as the structures of any intermediates formed.



[4]

- (d) When ester **A** ($\text{C}_6\text{H}_{11}\text{O}_2\text{Cl}$) is heated with alkaline KMnO_4 , two products are formed. Acidification of the two products give **B** and **C**.

C gives a yellow precipitate when warmed with aqueous alkaline iodine. When sodium borohydride is added to **C**, compound **D** is formed. **D** reacts with hot concentrated sulfuric acid to form **E** ($\text{C}_4\text{H}_6\text{O}_2$) that exhibits cis-trans isomerism.

B, **C**, **D** and **E** all produce effervescence when reacted with aqueous Na_2CO_3 .

Draw the structures of **A**, **B**, **C**, **D** and **E**.

[5]

[Total: 20]

END